2025 International Conference on Advancements in Power, Communication and Intelligent Systems (APCI) | 979-8-3315-2387-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/APCI65531.2025.11137103

Summarization and Translation of NPTEL Videos into Regional Indian Languages

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***Abstract*—In order to improve educational content accessibility for a variety of audiences, this study proposes a system for summarizing, translating, and turning NPTEL lecture transcripts into audio in regional Indian languages. Transcripts are preprocessed, succinct summaries are produced using sophisticated models like DistilBART, BART, T5, Google Pegasus, and BERTsum, and they are assessed using BLEU and ROUGE metrics. DistilBART outperformed the others with the highest BLEU score 45.13, ROUGE 1 F1, score of 74.31, and ROUGE 2 F1 score of 68.29 thereby showing competency in summarizing educational content. Technical words were kept in English for consistency while summaries were translated into Tamil, Telugu, Malayalam, Kannada, and Hindi using the Google API. Google Text-to-Speech (TTS) was used to turn the translated summaries into audio, resulting in outputs that were clear and natural and evaluated by native speakers and subject matter experts. By bridging linguistic and technological divides, this scalable multimodal solution makes excellent educational materials accessible to audiences that do not speak English.**

***Index Terms*—NPTEL, lecture summarization, machine translation, educational content, Indian languages, transcript summarization, text-to-audio.**

I. Introduction

Access to top-notch academic content has greatly increased thanks to online learning platforms like NPTEL. However, students may find the sheer volume and duration of these lecture videos daunting, particularly if they are looking for brief explanations of particular subjects. A further layer of complexity is added by India's linguistic variety, as many pupils find it difficult to access English-language content. By creating a system that automatically condenses NPTEL lecture videos and translates the condensed content into regional Indian languages, this project seeks to address these issues.

This paper aims to demonstrate that using specific summarization methods, aimless long pieces of text can be summarized and presented in brief, easy to read summaries. While using extractive models like TF-IDF and Gensim, Kulkarni et al. [1] were successful in minimizing content length and preserving key concepts successfully. Similarly to this, Chadawar et al. [7] explain how text summary and video processing can enhance students' attention by giving them important information in a shorter and more condensed form. We build upon these methods by proposing that NPTEL lecture transcripts should be summarized using both extractive and abstractive approaches. As mentioned by Mridha et al. [5], these models benefit from advancements in deep learning including RNNs and transformers which improve approach towards summary generation by taking into account contextual nuances.

Translation, however, appears to be a necessity in addition to summarization in this activity. It is important to translate these summaries into various local languages because of the multilingual setting in the Indian market. According to the authors Shweta Kalyanshetti et al, there are challenges that NMT models like transformers face including gender specific grammar and sentence structure differences in languages such as Marathi. The important role of effective translation models decreasing the linguistic disparities in the multilingual society was also highlighted by Gandhi et al. [4]. The purpose of this work is to launch this project to enhance students' access to the educational content available in NPTEL through translation and synopsis. It will also enable student to engage with the

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material in their languages thus enhancing the learning process in general.The proposed work helps to enhance the utility of NPTEL lecture videos in the following ways containing the following major elements.

- Transcription of NPTEL lecture videos into text through the Whisper model.
- Using models such as BERTSum, BART, T5, Pegasus, and DistilBART, the task is summarized with DistilBART bagging the best score.
- Translating the summaries into Tamil, Telugu, Malayalam, Kannada, Hindi, retaining technical terms in English language.
- Explaining how Google Text-to-Speech was used to download the translated summaries in an audio format for the auditory students.
- Guarantee retention of technical terms that would create readable and sound descriptions of summaries.

This makes NPTEL's excellent materials more accessible to students throughout India, regardless of language or location, this endeavor supports the larger objective of democratizing education.

The structure of the paper is organized as follows: Section 2 introduces related work, where the authors focus on summarization and translation in educational materials. Section 3 explains the approach, models and algorithms employed in the study. Section 4 explains the experimental results, while section 5 concludes the findings of the study and discusses directions for future research.

II. Related work

There are possibilities to use extractive and abstractive summarization techniques in textual and video document analysis. Thus, to achieve an efficient compression in presented evaluation indicators, Kulkarni et al. [1] and Awasthi et al. [8] focus on traditional extractive approach such as TF-IDF, Gensim and unsupervised clustering. These strategies, in turn, underlie the development of scalable summarization systems. Information sources have now evolved from simple pull models to more complicated architectures such as RNNs and transducers, as has been pointed out by Mridha et al. [5].

Integrated into the summarization pipeline are the contributions of Negro et al. [2], who designed off-line and privacy-orientated speech transcription with impressive results even in non-native speech environments. Building on this, Chadawar et al. [7] integrated transcription with summarization as well as keyframe identification, including LexRank, to improve students' attention in relevant education domains.

The data provided by Gandhi et al. [4] and Kalyanshetti et al. [6] refer to the fact that NMT is especially useful in filling gaps in linguistic communication. They also propose models based on hybrids and transformers for translating from Gujarati to English and Marathi. These models serve the need to handle syntactic and semantic incongruities and support multilingual applications in educational settings and beyond. Together, it is possible to highlight the role of NLP, video summarization, and translation technologies in determining effective and impactful educational content for various groups of learners.

III. Proposed Methodology

The proposed work uses multiple approaches for transcription, text summarization, translation, and text-to-speech conversion. Since NPTEL provides lecture videos, Whisper, a strong automatic speech recognition (ASR) model, is utilised for captioning. For the task of summarising, both extractive and abstractive models are used. BERTSum and BART models are used for extractive summarization, and in BERTSum, important sentences are identified considering the transformer-based architecture of BERT. T5 and Google Pegasus are used with abstractive summarization and Pegasus has further been fine-tuned as a result generate Summary. The accelerated model of BART: DistilBART generates the best summaries that are both concise and accurate at a reasonable computational cost. To translate the summaries into regional languages, the Google Translation API is used, and technical terms in Tamil, Telugu, Malayalam, Kannada, and Hindi are retained in English for better understanding. Following this, Google Text-to-Speech (TTS) is used to translate text into audio format. Dataset collection, pre-processing, transcript extraction, summarization, translation, text-to-audio, and comparison are some of the phases that make up this work (see Figure 1).

A. Dataset Description

The dataset for this research consists of 35 YouTube videos of NPTEL lectures on AI. These videos cover various AI topics and are the foundation for generating summary and translations, ensuring a diverse set of educational content to evaluate the effectiveness of the summarization and translation models. Figures 2 and 3 represent the statistics of the data set.

B. Dataset Pre-Processing

The preprocessing stage comprises of several important steps before the lecture material can be prepared for analysis. We get a standardised text written in lowercase, where punctuations are added, stopwords like 'the', 'is', etc. are removed. Finally, to prepare the text for summarization and translation, a Named Entity Recognition (NER) model is used to identify important entities such as persons, organizations, and locations. Additionally, Part-of-Speech (POS) tagging is applied to recognise the role of each word in the text.

C. Transcript Extraction

Automatic Speech Recognition (ASR) technologies are used to extract transcripts from NPTEL course videos. These methods ensured that technical words and context were maintained while accurately translating spoken speech into text. Excess noise was eliminated and the transcripts were carefully checked to make sure they matched the spoken material.

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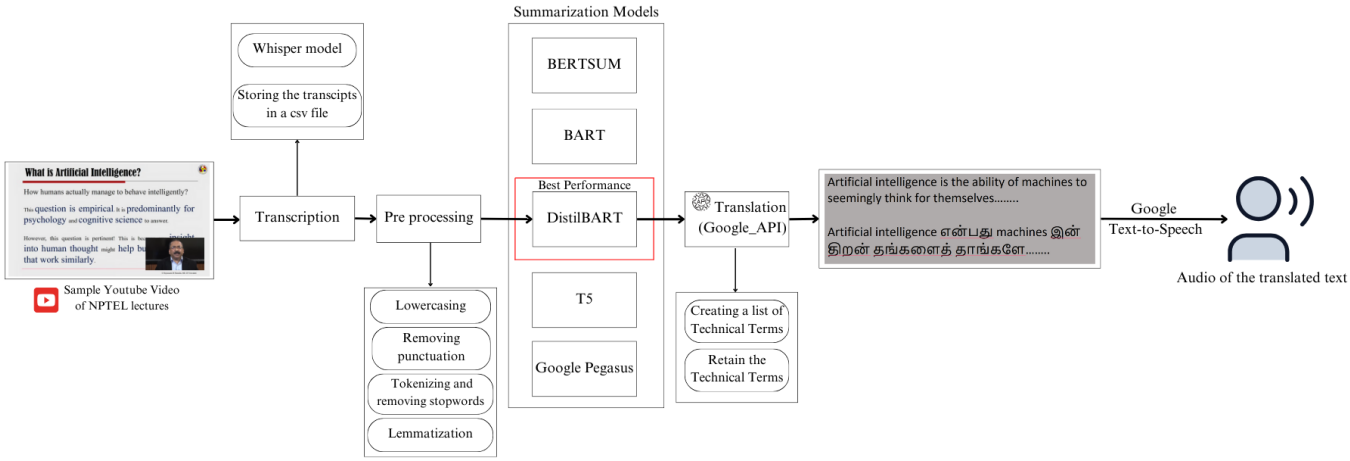

Fig. 1: Detailed Flow of Work

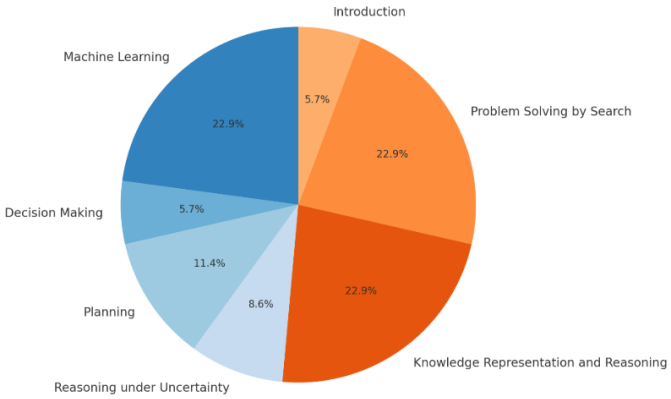

Fig. 2: Number of videos on each topic covered in the playlist

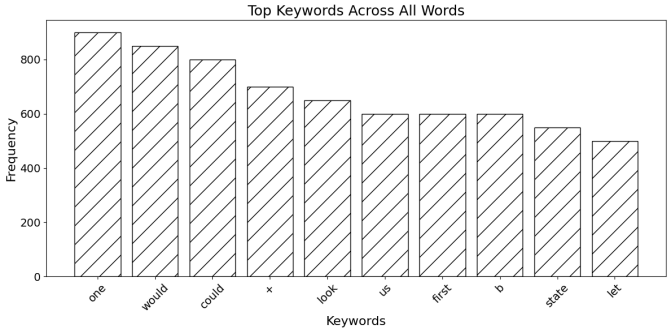

Fig. 3: Frequent Keywords in all NPTEL transcripts

D. Summarization

The recovered transcripts were fed into several summarizing models to generate concise and informative lecture summaries. Techniques for extractive and abstractive summarization were applied. Although extractive models chose key sentences from the transcript while keeping the original phrasing, abstractive models used semantic information to reword and condense the material. The objective was to produce summaries that successfully conveyed the main concepts and important details from the lecture material while maintaining the text's accuracy and coherence for instructional purposes.

E. Translation

Google API is used to translate the summaries into regional Indian languages: Hindi, Tamil, Telugu, Malayalam, and Kannada; since, as [11] points out, it yields reliable and accurate results for the majority of Indian languages. Since many technical words are well known and understood in the context of AI, their original English forms were retained to maintain technical integrity. "Artificial intelligence," "machine learning," "neural networks," "NLP," "deep learning," "CNN," "LSTM," and numerous other terminologies associated with AI, machine learning, and data science are included in the list of technical terms. It is made sure that the translated summaries have retained their technical correctness and applicability while allowing learners to access the content in their local tongues by preserving these terms. Google Text-to-Speech (TTS) is used to parse the translated summaries and convert the text into audio in the local language. The TTS settings are adjusted to ensure natural pronunciation, suitable speed, and clarity for instructional content.

F. Comparative Analysis

T5, BART, Google Pegasus, BertSum, and Distil BART for summarization models for NPTEL course transcripts were evaluated and for which short summaries were generated. The transcripts were then filtered to remove noise to fit into the input format of each model used in the study. These generated summaries were compared by using the standard related evaluations methods like BLEU which proposes the fluency and n-gram precision view and ROUGE providing the coherence and coverage of topics. To enhance the robustness of the evaluation, the METEOR which measures the quality of the summary by comparing synonyms with matching stems

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and semantic similarities and cosine similarity methods which compares alignment of word embeding between the generated summary and reference text were also employed.

IV. Results & Discussion

The evaluation of summarization models involves the use of multiple measures to provide a comprehensive assessment of the models. The measures used were BLEU, ROUGE (ROUGE-1 F1 and ROUGE-L F1), METEOR, and COSINE SIMILARITY metrics were used. All metrics point to specific aspects of the summarization quality and give a fairly fair estimate of the models.

Besides text summarization, the summaries were also translated into Tamil, Telugu, Malayalam, Kannada and Hindi with reasonably good contextual fidelity, primarily in retaining the referential value of the technical terms. As a result of possible confusion resulting from translating technical terms into local language adaptations, these terms were retained in their English terms. For example, terms such as 'Neural Network' and 'Back propagation', which should actually be translated, were kept in their English version, and this is because the authors wanted to keep it uniform around the world. However, the performance of the translations could not be calculated as no existing dataset was used and thus no reference translations were available. However, in the present study, Google API was used for the translation process, as suggested in [11], where the results showed great consistency between the various local languages used.

A fair comparison was not possible between models used in different work due to the usage of different data sets and other factors. The comparison of summarization models is summarised in Table I, in which specific information concerning the metrics that were used in the respective studies is presented. The table categorises models like BERT, RoBERTa, T5, BART, and Primera, Pegasus, LED etc., and different datasets like Multi-News, Multi-XScience, PubMed etc. This work further extends the evaluation of the framework with NPTEL course transcripts using improved models such as DistilBART with the highest BLEU score 45.13, ROUGE 1 F1, score of 74.31, and ROUGE 2 F1 score of 68.29, thus showing competence in summarising educational content. BART performed the next best with a BLEU score of 41.66 and high ROUGE values indicating a favourable trade-off between correctly generated information and using as few rephrased words as possible. These results demonstrate the effectiveness of DistilBART and BART in producing fluent and semantically accurate summaries supported by translation of the summaries into multiple languages and conversion of the summaries into speech for wider accessibility.

Figure 4 shows the five summarization models: DistilBART, BART, T5, Google Pegasus, and BERTSum, in different evaluation matrices. All have been chosen since each of them reveals a different fact of the quality of the summaries produced and also provides a comprehensive summary of the performance of various summarization models based on multiple evaluation metrics: The evaluation metrics used are Bilingual Evaluation Understudy (BLEU), Reciprocal n-gram Overlap as Generated by ROUGE version 1.5.5 F1, Longest ROUGE n-gram F1, Meteor, and Cosine Similarity. The results of each of the models in each of these metrics are presented to show how effective each of the models is in creating summaries.

- DistilBART: Achieved the highest scores on all aspects under evaluation and, therefore, was able to produce highly accurate, fluent, and contextually relevant summaries.
- BART: BART performed neatly and was very similar to DistilBART. It registered the highest METEOR of (36.00) thus emphasising the ability of the tool to match meanings optimally.
- T5: Competitive performance was achieved in both tasks, specifically in BLEU and ROUGE scores which indicates its ability to uphold context and its ability to create immense depth summaries.
- Google Pegasus: Although it was trained for abstractive summarization, the numbers on the whole were somewhat lower, especially in aspects to do with ROUGE-L F1, suggesting the model needs to be better at organising a coherent summary.
- BERTSum: Largely tested for extractive form, it yielded moderate scores on ROUGE but struggled on BLEU and METEOR suggesting that it is a less efficient system for abstracting summaries.

Besides text summarization, the summaries were also translated into Tamil, Telugu, Malayalam, Kannada, and Hindi with reasonably good contextual fidelity, primarily in retaining the referential value of the technical terms. Translation of a sentence from the summaries is shown in figure 5. As a result of possible confusion resulting from translating technical terms into local language adaptations, these terms were retained in their English terms. For instance, the terms like "Neural Network" and "Back propagation" which should actually be translated were kept in their English version and that is for the reason that the authors wanted to keep it uniform around the world. However, the performance of the translations could not be calculated as no existing dataset was used and thus no reference translations were available. However, in the present study, the Google API was used for the translation process, as suggested in [11], where the results showed great consistency between the various local languages used.

The translated text was read aloud using Google Text- to-Speech (TTS) to convert the text into speech as well. According to the authors, the naturalness and clarity of the ensuing audio outputs are high. Both the different translations and the corresponding audio renditions were checked with domain specialists and native speakers for accuracy as well as naturalness of the displayed translations. The last multimodal output is audio summaries in the localized language, which enhance access to instructional information. This is particularly helpful to a valued audience who has no English

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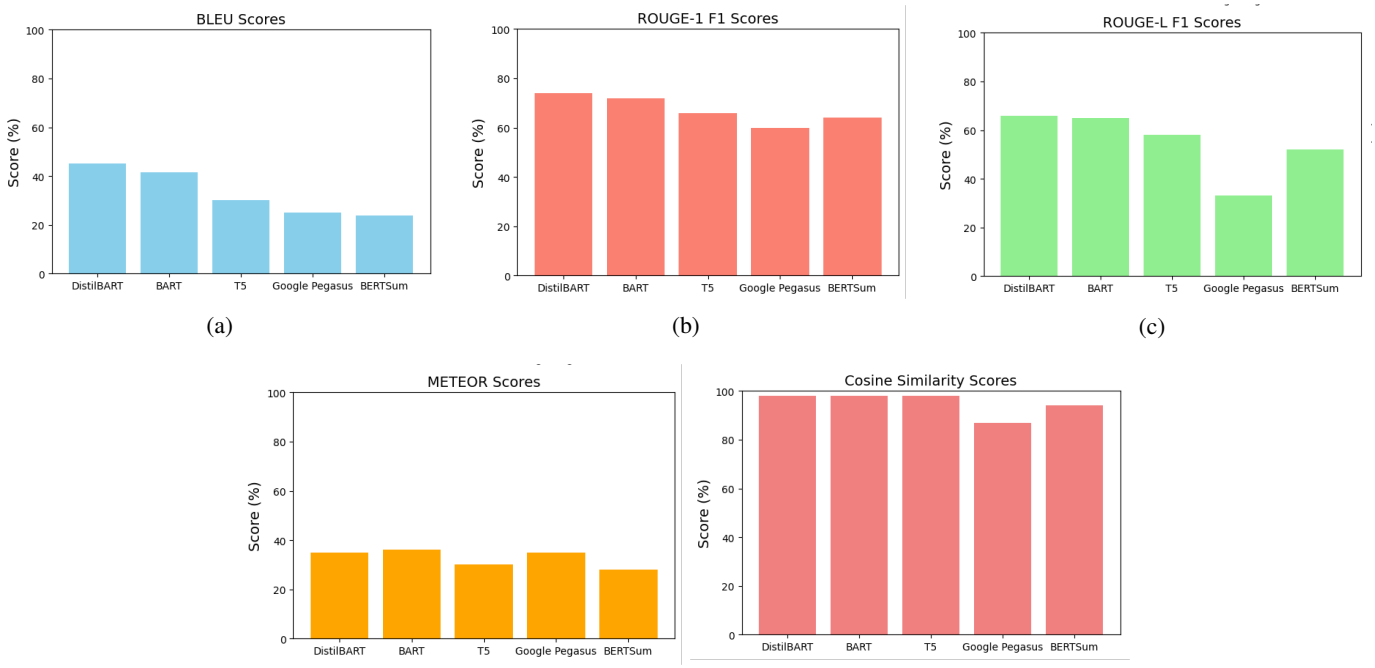

(a) (b) (c) (d) (e)

Fig. 4: Evaluation Scores for Different Metrics: (a) BLEU Scores, (b) ROUGE-1 F1 Scores, (c) ROUGE-L F1 Scores, (d) METEOR Scores, (e) Cosine Similarity Scores

TABLE I: Model Performance Comparison

Model Used	Year	Dataset Used	ROUGE - 1	ROUGE - 2	BLEU
BERT [13]	2022	Private Dataset	40	31	48
RoBERTa [13]			37	23	38
T5 [13]			36	25	54
BART [13]			58	46	56
Primera [18]	2023	Multi-News	42	13.6	-
		Multi-XScience	29.1	4.6	
Pegasus [18]		Multi-News	32	10.1	-
		Multi-XScience	27.6	4.6	
LED [18]		Multi-News	17.3	3.7	-
		Multi-XScience	14.6	1.9	
Discourse-Aware Summarization [12]	2024	PubMed	38.93	15.37	-
BARTchunk with a pre-trained large CNN [12]		PubMed	40.74	14.77	-
		CNN/Daily Mail	43.02	20.57	
BARTchunk with a pre-trained large XSum [12]		CNN/Daily Mail	28.47	8.07	-
DistilBART (Model Used in This Work)	**2024**	**NPTEL Course Transcripts**	**74.31**	**69.89**	**45.13**
BART (Model Used in This Work)			**72.09**	**68.29**	**41.66**
T5 (Model Used in This Work)			**66.16**	**60.51**	**30.23**
Google Pegasus (Model Used in This Work)			**59.88**	**30.71**	**25.07**
BERTSum (Model Used in This Work)			**64.00**	**53.46**	**23.90**

skills in comprehending educational content, hence promoting diversity.

V. Conclusion

This study effectively illustrated how to summarise NPTEL course transcripts, translate the summaries into local languages, and use Google Text-to-Speech (TTS) to create

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Text to be translated: Artificial intelligence is the ability of machines to seemingly think for themselves.
Translated text(Hindi): Artificial intelligence machines की क्षमता है जो खुद के लिए लगता है।
Translated text(Malayalam): Artificial intelligence സ്വയം ചിന്തിക്കാനുള്ള machines കഴിവാണ്.
Translated text(Kannada): Artificial intelligence machines ನ ಸಾಮರ್ಥ್ಯವು ತಮ್ಮನ್ನು ತಾವು ಯೋಚಿಸುವ ಸಾಮರ್ಥ್ಯ.
Translated text(Telugu): Artificial intelligence అంటే machines సామర్థ్యం తమను తాముగా ఆలోచించడం.
Translated text(Tamil): Artificial intelligence என்பது machines இன் திறன் தங்களைத் தாங்களே சிந்திக்கத் தோன்றுகிறது.

Fig. 5: Translation of sample texts in different languages

the matching audio. With the highest BLEU score (45.13), ROUGE-1 F1 (74.31), and ROUGE-L F1 (66.23) among the summarising models tested, DistilBART demonstrated its ability to provide the most fluid and contextually accurate summary. With a BLEU score of 41.66 and high ratings of ROUGE-1 F1 (72.09) and ROUGE-L F1 (65.00), BART came second. The informativeness and shortness of these two models were consistently balanced. With a BLEU score of 30.23 and a METEOR score of 36.00, T5 demonstrated a moderate level of performance, indicating its ability to preserve semantic fidelity. Despite having somewhat lower BLEU and ROUGE scores, Google Pegasus and BERTSum produced outputs that were coherent; BERTSum maintained a cosine similarity of 94.00, indicating good alignment with reference texts. In particular, all models obtained high cosine similarity scores of 98.10, indicating strong semantic agreement with reference summaries, with the exception of DistilBART (cosine similarity: 87.21). Using the Google API, summaries were translated into Tamil, Telugu, Malayalam, Kannada, and Hindi while maintaining contextual fidelity. To ensure understanding for audiences who do not know English, technical terms were kept continuously in English. Google TTS improved accessibility for a variety of learners by producing clear and natural audio outputs. This study provided a detailed evaluation of the summarization models by combining several evaluation measures, including BLEU, ROUGE, METEOR, and cosine similarity. The results demonstrate the system's ability to overcome linguistic and technological obstacles and identify DistilBART and BART as the best models for producing accurate, contextually relevant, and fluid summaries. This multimodal method promotes inclusivity in education by greatly increasing the access of regional language speakers to high-quality educational information.

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